

FLEXBOX (Lecture 13)

Coding & Web Development | Nishant Sir | Summary Notes - structured & detailed

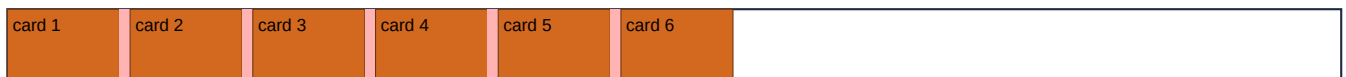
1. Flexbox kya hai?

Flexbox = Flex (flexible) + Box. Ye CSS ka ek **layout model** hai jo ek container ke andar ke items ko ek line (row ya column) me **arrange, align aur space-distribute** karta hai - bina float / inline-block / margin hacks ke. Container ki size change ho (responsive), items khud adjust ho jaate hain.

Term	Matlab	Example
Flex container	Parent element jisme <code>display: flex</code> lagta hai	<code>.container { display: flex; }</code>
Flex items	Container ke <u>direct children</u> - sirf wahi flex hote hain (grandchildren nahi)	card 1 ... card 6
Main axis	Items jis direction me chalte hain. Default = left to right (row). <code>justify-content</code> isi par kaam karta hai	horizontal ->
Cross axis	Main axis ke perpendicular (default top to bottom). <code>align-items</code> isi par kaam karta hai	vertical

2. Flexbox se pehle ki problem: inline-block ka whitespace gap

Cards ko ek line me laane ke liye pehle `display: inline-block` use hota tha. Problem: cards ke beech **chhota sa gap** aa jaata hai (diagram me red).



red = whitespace gap (~4px)

- **Reason:** HTML me do tags ke beech jo **space / Enter (new line)** hota hai, browser use ek **text space character** maanta hai. *inline-block* elements text ki tarah inline flow me baithte hain, isliye ye space render hota hai (~4px, font-size par depend).
- **Fixes (purane):** (a) parent par `font-size: 0;` (b) tags ke beech ka whitespace hata do (ek hi line me likho / HTML comment lagao); (c) negative margin.
- **Best fix = Flexbox:** flex items **blockified** ho jaate hain - unke beech ka whitespace **ignore** hota hai, to unwanted gap hi nahi aata.

3. Flexbox shuru kaise karein

```
.container {  
  display: flex;           /* items ab ek row me aa gaye */  
  justify-content: space-around; /* main axis par spacing */  
}  
.card { width: 100px; height: 100px; background: chocolate; }
```

- Sirf `display: flex` likhte hi: items row me aate hain, `flex-direction: row` default hota hai, aur items apni content-width jitni jagah lete hain (stretch nahi hote jab tak width na ho).
- Total **free space** = container width - (saare items ki width ka sum). `justify-content` isi free space ko **distribute** karta hai.

4. justify-content - main axis par spacing

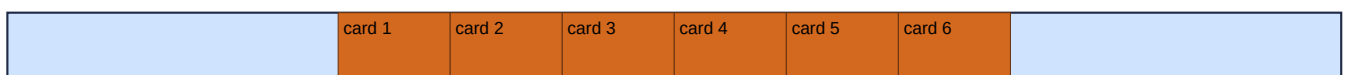
Example setup: container **W**, **n = 6** cards, free space = **F**. Neeche har value ka diagram + formula.

justify-content: flex-start (default); - Sab items shuru (left) me, free space end me.



free space sab right me (light-blue = free space)

justify-content: center; - Items beech me; free space dono side barabar ($F/2 + F/2$).



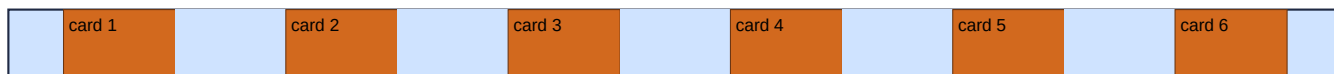
free space dono side barabar (light-blue = free space)

justify-content: flex-end; - Sab items end (right) me, free space shuru me.



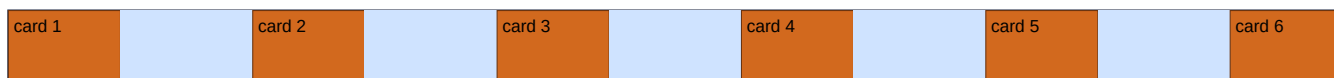
free space sab left me (light-blue = free space)

justify-content: space-around; - Har item ke dono side barabar space (F/n ka aadha-aadha). **Edge gap = beech ke gap ka aadha.**



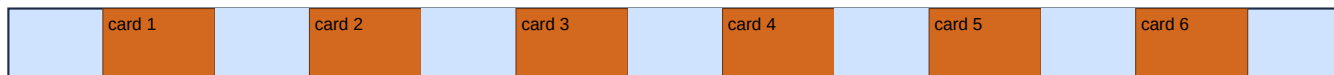
$edge = x/2$ | $beech = x$ (har item ke dono side $x/2$) (light-blue = free space)

justify-content: space-between; - Pehla item bilkul start par, aakhri bilkul end par; **beech ke (n-1) gaps barabar.** Edge gap = 0.



$edge = 0$ | $beech me barabar gap$ (light-blue = free space)

justify-content: space-evenly; - **Saare gaps barabar** - edges ke bhi aur beech ke bhi. Total gaps = $n+1$.



$edge = beech = same gap (n+1 gaps)$ (light-blue = free space)

5. Gap ka math (concrete example) - exam me kaam aayega

Maan lo container = **600px**, 6 cards $\times$ **60px** = 360px. Free space **F = 240px**.

Value	Formula	Edge gap (left/right)	Beech ka gap
space-around	each item ko $F/n = 240/6 = 40$ mile, dono side 20-20	20px	$20+20 = \mathbf{40px}$
space-between	$F/(n-1) = 240/5$	0px	48px
space-evenly	$F/(n+1) = 240/7$	~34.3px	~34.3px (same)

6. Quick comparison & yaad rakhne ki tricks

Property	Edges par space?	Beech ka space	Yaad rakhne ka trick
space-around	Haan - aadha	Edge ka double	"Around" = har item ke around barabar jagah; do items ke beech 2 hisse mil jaate hain
space-between	Nahi	Barabar	Space sirf items ke between
space-evenly	Haan - full	Edge ke barabar	Evenly = har jagah same, koi favourite nahi

7. Common mistakes & revision points

- **justify-content parent par lagta hai**, items par nahi. Items par lagaya to kuch nahi hoga.
- Agar items container ki poori width le lete hain (free space = 0), to *space-around/between/evenly* ka koi visible effect nahi dikhega.
- **Default flex-direction = row**, to *justify-content* horizontal chalta hai. *flex-direction*: column karte hi main axis vertical ho jaata hai aur *justify-content* bhi vertical ho jaata hai.
- Sirf **direct children** flex items bante hain; nested elements ko alag se `display: flex` dena padta hai.
- inline-block ka gap **bug nahi**, `whitespace-as-text` ka natural result hai; flexbox me ye issue hi nahi hota.
- **Axis yaad rakho**: *justify-content* = main axis, *align-items* = cross axis.